

Surgical Safety QA

Templates B1–B7 • Annotation-first generation

53 validated question-answer pairs, organized by template

July 29, 2026

Corpus overview

Each entry uses the exact template question and reports video time in minutes and seconds. Model reasoning was required and verified during generation; private thinking traces are not reproduced here.

Template	Purpose	Selected sources
B1 • 9	Dangerous Frame Critique	BernBypass70 (3), Endoscapes2023 (3), StrasBypass70 (3)
B2 • 10	Is It Safe To Proceed?	CholecT50 (5), Endoscapes2023 (5)
B3 • 9	Risk Ranking	BernBypass70 (3), Endoscapes2023 (3), StrasBypass70 (3)
B4 • 6	Near-Miss Detection	BernBypass70 (3), StrasBypass70 (3)
B5 • 9	Stop Point	BernBypass70 (3), Endoscapes2023 (3), StrasBypass70 (3)
B6 • 5	Cause of Error	MultiBypass140 (5)
B7 • 5	CVS Safety Check	Endoscapes2023 (5)

Selection note. B1, B3, and B5 use three distinct videos from each of BernBypass70, StrasBypass70, and Endoscapes2023. B4 uses six deliberately difficult negative controls from the two bypass cohorts: the source labels document actual adverse events, so they must not be mislabeled as near misses. CholecT50 was not used for B1/B3/B4/B5 because its local annotations do not identify safety events.

B1 | Dangerous Frame Critique

9 QA pairs • *BernBypass70*: 3, *Endoscapes2023*: 3, *StrasBypass70*: 3

B1.1 • *BernBypass70* • BBP01 • 23:43-23:44

Question: At timestamp 23:43-23:44, identify the most urgent patient-safety concern.

Answer: The ongoing thermal injury to the dissected crural tissue is the most urgent safety issue right now. Unmanaged heat transfer in this region creates an immediate risk of extending damage into adjacent structures, so you need to pause all energy application. Carefully withdraw the instrument, allow the tissue to cool, and reassess the dissection plane before proceeding. If further hemostasis or dissection is required, switch to cold techniques or use precise, low-power coagulation with direct visualization to keep the thermal field strictly within the intended boundaries.

Rationale: The available annotations for this video span explicitly document a thermal injury event during the gastric pouch creation phase, specifically within the crura dissection step. The event is recorded as unrectified, indicating that the tissue damage is ongoing or unresolved, which elevates the clinical urgency. On the provided severity scale, a rating of 3 reflects a moderate-to-high level of tissue compromise that demands prompt intervention to prevent progression to adjacent structures. Because the annotation scope limits itself to event categorization and timing without describing visual appearance or physical cause, the coaching focuses on the standard clinical management of unrectified thermal injuries in this anatomical region. The guidance emphasizes halting energy application, allowing tissue recovery, and utilizing alternative dissection methods to safeguard the crura and neighboring anatomy.

B1.2 • *BernBypass70* • BBP39 • 14:41-14:45

Question: At timestamp 14:41-14:45, identify the most urgent patient-safety concern.

Answer: The most urgent concern is a severe, unrectified mechanical injury occurring during the gastrojejunal stapling phase. This event places the fresh anastomosis and surrounding gastrointestinal tissues at immediate risk of perforation, bleeding, or structural compromise. Because the injury has not been resolved, you must halt the stapling maneuver right away, maintain careful exposure of the operative field to evaluate the extent of the damage, and take immediate steps to control the mechanical breach before any further progression is safe.

Rationale: The answer is derived directly from the provided annotations. The `recorded_event_category` specifies a 'Mechanical injury', which forms the core safety concern. The `recorded_event_severity` of 5 indicates the maximum level of severity, which is translated into the highest clinical urgency and the descriptor 'severe'. The `recorded_event_rectified` status of False confirms the injury is ongoing and unaddressed. The `operative_phase` 'gastrojejunal_anastomosis' and `operative_step` 's18_gastrojejunal_stapling' supply the explicit anatomical and procedural context used to locate the risk. The corrective guidance focuses on halting the stapling and assessing the breach, which is standard clinical management for this type of adverse event. The rationale notes that visual appearance and physical cause are explicitly excluded from the annotation scope, preventing any invented details.

B1.3 • BernBypass70 • BBP54 • 74:32-79:53

Question: At timestamp 74:32-79:53, identify the most urgent patient-safety concern.

Answer: The most urgent patient-safety concern at this timestamp is the incomplete closure of the gastrojejunal anastomosis. Because the surgical site remains open and unsecured, there is a direct risk of leakage and subsequent intra-abdominal infection. Immediate corrective action is required to properly reinforce or revise the anastomotic closure before proceeding.

Rationale: The answer is grounded in the BernBypass70 annotations for video BBP54, clip b1_bern_03. The provided metadata explicitly records an adverse event categorized as 'Insufficient closure of anastomosis' during the gastrojejunal anastomosis phase and the s19_gastrojejunal_defect_closing step. The event carries a severity rating of 1 on a 1-to-5 scale, indicating a low-severity event, but it is marked as unrectified. This combination signals a documented but baseline-level concern that requires prompt correction to prevent downstream complications. The annotation scope explicitly states that the labels establish the occurrence of the event but do not describe its visual appearance or physical cause, so the response strictly adheres to the clinical implications of an unsecured anastomosis without inventing procedural details.

B1.4 • Endoscapes2023 • 57 • 18:26

Question: At timestamp 18:26, identify the most urgent patient-safety concern.

Answer: The operative field currently lacks the clear anatomical exposure required to safely proceed. Because the hepatocystic triangle remains uncleared and the cystic duct and artery are not distinctly visualized, proceeding with clipping or cutting carries a significant risk of inadvertently injuring the common bile duct or major hepatic vessels. You should pause any definitive dissection or device deployment until the triangle is fully cleared and both the cystic duct and artery are confirmed separate from surrounding tissue.

Rationale: The Endoscapes2023 annotations for this timestamp show that the Critical View of Safety criteria for structure identification (cvs_structures_identified) and hepatocystic triangle clearance (cvs_hepatocystic_triangle_cleared) are both recorded at 0.00, indicating they are unmet. The criterion for gallbladder separation (cvs_gallbladder_separated) is 1.00. Following the conservative prerequisite order for surgical safety, structure identification must be achieved first, followed by triangle clearance. The labeled structures present (cystic artery, cystic duct, gallbladder) confirm the relevant anatomy, but the zero achievement for the first two prerequisites establishes a clear safety-readiness gap. The urgent concern is the lack of confirmed isolation, which directly increases the risk of biliary or vascular injury if dissection or clipping proceeds. No discrete adverse event is recorded, so the feedback focuses on preoperative preparation rather than rectifying an active complication.

B1.5 • Endoscapes2023 • 74 • 8:02

Question: At timestamp 8:02, identify the most urgent patient-safety concern.

Answer: Incomplete identification of the critical anatomical structures is the immediate safety priority right now. Before any clipping or cutting can be considered safe, you must ensure complete and unambiguous visualization of the required structures to prevent accidental injury to the biliary tree or vasculature.

Rationale: The provided annotations for timestamp 8:02 indicate that the Critical View of Safety criterion for structure identification has a recorded value of 0.67, meaning it has been achieved by a majority but not

unanimously, while the criteria for hepatocystic triangle clearance and gallbladder separation are fully achieved at 1.00. Following the safety-readiness framework, the partially met structure identification criterion is the most urgent unmet prerequisite. No discrete adverse event or visible injury is documented at this timestamp; the concern is strictly one of anatomical exposure and readiness to proceed. The rationale relies solely on the provided CVS criterion scores and labeled structure classes.

B1.6 • Endoscapes2023 • 8 • 11:51

Question: At timestamp 11:51, identify the most urgent patient-safety concern.

Answer: The immediate safety concern is the absence of the anatomical exposure required to safely proceed with clipping or cutting. Attempting to divide the cystic duct or artery without a definitive, unambiguous view of the hepatocystic triangle and the gallbladder-liver interface risks iatrogenic injury to the bile duct or major vessels. The corrective step is to pause all dissection and clipping, and focus entirely on achieving clear, direct visualization of the cystic duct, cystic artery, and the gallbladder bed before advancing.

Rationale: The input provides Critical View of Safety criterion scores for structure identification, hepatocystic triangle clearance, and gallbladder separation, all recorded as 0.00, indicating non-achievement. Per the conservative prerequisite hierarchy for safety readiness, when multiple CVS criteria are equally unmet, the highest priority is the failure to clearly identify the required anatomical structures before attempting to clear the triangle or separate the gallbladder. The presence of labeled tool and structure classes confirms the operative context but does not override the safety prerequisite for unambiguous anatomical identification. The answer translates this unmet criterion into direct clinical guidance regarding the risk of proceeding without clear exposure and the need to pause for anatomical confirmation.

B1.7 • StrasBypass70 • SBP06 • 68:32-71:04

Question: At timestamp 68:32-71:04, identify the most urgent patient-safety concern.

Answer: A severe, uncorrected mechanical injury to the jejunal mesentery is the immediate priority right now. Because this damage has not been addressed, there is a high risk of ongoing tissue compromise, bleeding, or ischemia in the affected bowel segment. You must stop the current maneuver immediately to assess the extent of the mesenteric tear, achieve hemostasis, and restore vascular integrity before continuing with the jejunal separation.

Rationale: The question asks for the most urgent patient-safety concern at the specified timestamp. The available annotations document a discrete intraoperative adverse event during the jejunal separation phase, specifically the mesenteric opening step. The event category is recorded as a mechanical injury, and its unrectified status combined with the highest severity band indicates a critical, unresolved tissue compromise that demands immediate attention. Because the provided metadata only confirms the occurrence and timing of this injury without describing its visual appearance or physical mechanism, the feedback focuses on the standard clinical response to an unaddressed mesenteric tear: halting the procedure to evaluate vascular integrity and control any bleeding. The rationale translates the raw labels (recorded_event_category: Mechanical injury, recorded_event_severity: 5, recorded_event_rectified: False) into direct clinical guidance without referencing the annotation system itself in the answer.

B1.8 • StrasBypass70 • SBP52 • 13:49-14:52

Question: At timestamp 13:49-14:52, identify the most urgent patient-safety concern.

Answer: Active bleeding during the dissection of the His angle while creating the gastric pouch is the most urgent concern right now. Because this hemorrhage has not been controlled, it poses a direct threat to patient safety by obscuring the surgical field and risking further tissue injury. You need to pause the dissection immediately to secure hemostasis and regain clear visualization before continuing with any further steps in the pouch creation.

Rationale: The input provides an event_timestamp of 13:49-14:52 with a recorded_event_category of Bleeding. The operative_phase is gastric_pouch_creation and the operative_step is s5_his_angle_dissection, which grounds the anatomical context. The recorded_event_severity is 4, indicating a high level of urgency on the dataset scale, and recorded_event_rectified is False, meaning the event is ongoing and unaddressed. These labels combine to establish active, uncontrolled bleeding as the primary safety issue requiring immediate attention. Per the event_scope_limit annotation, the provided metadata confirms the event's occurrence but does not supply visual details of the bleeding's appearance or physical cause, so the answer relies strictly on the categorical and contextual labels.

B1.9 • StrasBypass70 • SBP63 • 22:48-26:37

Question: At timestamp 22:48-26:37, identify the most urgent patient-safety concern.

Answer: Proceeding without addressing the active thermal damage to the gastric tissue during pouch creation is the primary safety priority right now. This uncorrected burn carries a substantial risk of tissue breakdown or leakage, which is why the procedure must be paused and the affected area carefully inspected. Before any further stapling or dissection occurs, the integrity of the stomach wall needs to be verified to prevent delayed perforation or postoperative complications.

Rationale: The question uses the exact B1 template wording with the provided timestamp. The answer is grounded in the StrasBypass70 annotations for video SBP63, clip b1_strasbourg_02. The recorded_event_category label identifies a thermal injury, and the operative_phase/operative_step labels place it during gastric_pouch_creation and s11_gastric_opening, allowing the specification of gastric tissue as the site at risk. The recorded_event_rectified label is marked False, indicating the injury has not been resolved, which elevates the urgency. The recorded_event_severity score of 3 (on a 1-5 scale where higher values indicate worse outcomes) translates clinically to a significant concern requiring immediate intervention to prevent tissue compromise. The event_scope_limit note confirms that no visual appearance or physical cause is provided, so the answer avoids describing specific tissue damage, instrument behavior, or the mechanism of injury beyond the supplied category. All raw field names, numeric values, and annotation terminology are restricted to this rationale per hard rule 11.

B2 | Is It Safe To Proceed?

10 QA pairs • CholecT50: 5, Endoscapes2023: 5

B2.1 • CholecT50 • VID01 • 16:50

Question: Based on the frame at 16:50, is it medically safe to proceed with coagulating the cystic artery with bipolar forceps? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. The procedural context indicates the intended maneuver is coagulation of the cystic artery using bipolar forceps during gallbladder dissection, but critical safety parameters remain unverified. Anatomical exposure of the target vessel cannot be confirmed, leaving the precise tissue plane and hemostasis status uncertain. The exact instrument position relative to the artery and adjacent structures is unknown, and the active state of the energy device cannot be verified. Proceeding without confirming these factors risks unintended thermal injury or vascular compromise.
[VISUAL EVIDENCE NEEDED]

Rationale: The annotations for this clip identify the surgical phase as gallbladder dissection and the intended action as coagulating the cystic artery with bipolar forceps. However, the B2 safety assessment requires verification of anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety, all of which are explicitly marked as unlabeled factors. Without frame-level visual confirmation of these parameters, it is impossible to determine whether the target structure is adequately exposed or isolated, whether hemostasis is achieved, whether the instrument is correctly positioned to avoid collateral damage, or whether the energy device is active and safe to use. The absence of these labeled safety checks means the procedure cannot be cleared for advancement at this timestamp.
[VISUAL EVIDENCE NEEDED]

B2.2 • CholecT50 • VID02 • 21:14

Question: Based on the frame at 21:14, is it medically safe to proceed with clipping the cystic duct? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. While the workflow context indicates that the clipper is engaged with the cystic duct during the clipping-and-cutting phase, the critical safety prerequisites for this maneuver cannot be verified from the available information. Specifically, there is no confirmed evidence of adequate anatomical exposure, the tissue plane has not been verified, hemostasis status is unknown, the precise instrument position relative to surrounding structures cannot be confirmed, and the energy device safety remains uncertain. Direct visualization of these factors is mandatory to ensure that the correct structures are isolated and that clipping proceeds without risk of bile duct injury or bleeding.
[VISUAL EVIDENCE NEEDED]

Rationale: The question asks for a safety check before clipping the cystic duct at timestamp 21:14. The available annotations indicate the surgical phase is clipping-and-cutting and the action triplet identifies a clipper engaging the cystic duct. However, the B2 template requires verification of anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety. The provided metadata explicitly lists these

as unlabeled B2 factors. Because these critical safety parameters are not recorded in the annotations, it is impossible to confirm that the necessary anatomical prerequisites are met or that the environment is safe for clipping. The answer relies on the absence of these labeled safety factors to recommend a pause for direct visual confirmation. [VISUAL EVIDENCE NEEDED]

B2.3 • CholecT50 • VID04 • 15:56

Question: Based on the frame at 15:56, is it medically safe to proceed with cutting the cystic duct? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. Before cutting the cystic duct, you must visually verify that the critical view of safety is fully established, with clear exposure of the hepatocystic triangle and two distinct structures entering the gallbladder. You also need to confirm that the tissue plane around the duct is clearly dissected and free of fibrous attachments, that hemostasis is secure with no active bleeding obscuring the field, that the tip of the scissors is positioned correctly along the intended cutting line without encroaching on adjacent anatomy, and that the device is in a safe state for activation. Because these critical safety conditions cannot be verified from the available information, you cannot safely proceed. [VISUAL EVIDENCE NEEDED]

Rationale: The provided annotations for timestamp 15:56 only supply an action triplet indicating scissors are cutting the cystic duct and place the procedure in the clipping-and-cutting phase. The input explicitly flags tissue plane, hemostasis, instrument position, and energy safety as unlabeled factors required for this safety assessment. Because the B2 template demands visual confirmation of anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety to justify proceeding, and these elements are absent from the annotation set, the evaluation identifies these critical gaps. The action label confirms the intended maneuver, but without visual evidence for the safety prerequisites, a definitive safety determination cannot be made. [VISUAL EVIDENCE NEEDED]

B2.4 • CholecT50 • VID05 • 15:27

Question: Based on the frame at 15:27, is it medically safe to proceed with dissecting the cystic plate with a hook? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. The current context indicates that a grasper is retracting the gallbladder and a hook is engaged with the cystic plate during a dissection phase, but the critical safety checks required before advancing cannot be verified. Specifically, the anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety remain uncertain. Proceeding without confirming these conditions risks inadvertent injury to adjacent structures or uncontrolled bleeding. [VISUAL EVIDENCE NEEDED]

Rationale: The available annotations identify the current action as 'grasper,retract,gallbladder' and 'hook,dissect,cystic_plate' within the 'carlot-triangle-dissection' phase. However, the B2 template requires explicit verification of anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety. The input explicitly marks these factors as unlabeled. Because the safety prerequisites for safe dissection

cannot be confirmed from the provided metadata alone, direct visual confirmation is mandatory before proceeding. [VISUAL EVIDENCE NEEDED]

B2.5 • CholecT50 • VID06 • 34:17

Question: Based on the frame at 34:17, is it medically safe to proceed with aspirating fluid with the irrigator? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. While the operative context indicates cleaning and coagulation with an irrigator actively aspirating fluid, verifying safe progression requires direct visualization of several critical safety parameters. You must confirm that the aspiration tip is correctly positioned within the intended tissue plane and anatomical exposure, ensure that surrounding structures are adequately retracted to prevent unintended contact, verify that hemostasis is maintained in the visual field, and confirm that any adjacent energy devices are safely deactivated or shielded. Without these direct visual checks, the risk of tissue trauma, fluid aspiration into unintended spaces, or thermal injury remains unquantified. [VISUAL EVIDENCE NEEDED]

Rationale: The request asks for a safety assessment at timestamp 34:17 for the action of aspirating fluid with an irrigator. The available annotations identify the surgical phase as cleaning-and-coagulation and provide action triplets indicating the irrigator is aspirating fluid alongside grasper actions on the specimen bag and liver retraction. However, the annotation set explicitly notes that the critical B2 safety factors (tissue plane, hemostasis, instrument position, and energy safety) are unlabeled. Because these specific safety parameters cannot be verified from the provided metadata alone, direct visual confirmation of the operative field is required to rule out risks such as inadvertent tissue contact, compromised hemostasis, or proximity to active energy sources. [VISUAL EVIDENCE NEEDED]

B2.6 • Endoscapes2023 • 12 • 17:28

Question: Based on the frame at 17:28, is it medically safe to proceed with continuing dissection toward clipping or cutting the cystic duct or artery? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. The foundational anatomical landmarks required to safely isolate the cystic duct and artery have not been adequately exposed or cleared, leaving the tissue plane undefined and critical structures unprotected. I cannot currently verify whether active bleeding is present, confirm the precise placement of the instruments near the target anatomy, or determine the status of any energy devices that may be in use.

[VISUAL EVIDENCE NEEDED]

Rationale: The requested safety judgment relies on multiple factors that are not fully captured in the available annotations. The Critical View of Safety criteria for structure identification, hepatocystic triangle clearance, and gallbladder separation are all recorded at 0.00, indicating that the necessary anatomical exposure has not been achieved. While the presence of the calot triangle, cystic artery, cystic duct, cystic plate, and gallbladder is noted, these class labels do not describe their condition or spatial relationships. Additionally, the annotations explicitly lack data on hemostasis, actual instrument position, and energy safety, which are

required to complete the B2 safety assessment. Without direct visual confirmation of these elements, a definitive safety determination cannot be made.

[VISUAL EVIDENCE NEEDED]

B2.7 • Endoscapes2023 • 15 • 19:06

Question: Based on the frame at 19:06, is it medically safe to proceed with continuing dissection toward clipping or cutting the cystic duct or artery? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. While the anatomical exposure and tissue planes required for the critical view are largely established, I cannot verify whether hemostasis is secure, confirm the exact instrument position relative to vulnerable structures, or assess energy safety. [VISUAL EVIDENCE NEEDED]

Rationale: The question asks whether it is safe to proceed with clipping or cutting based on visible evidence for anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety. The available annotations provide Critical View of Safety scores for structure identification (0.67), hepatocystic triangle clearance (0.67), and gallbladder separation (1.00), along with a list of present structure classes. While these metrics indicate that the structural prerequisites for the critical view are largely met, the provided metadata explicitly notes that hemostasis, actual instrument position, and energy safety are unlabeled B2 factors. Because the critical view criteria and class-presence labels do not cover hemostasis, precise instrument placement, or energy safety, the full safety judgment required by the prompt cannot be completed from the annotations alone. [VISUAL EVIDENCE NEEDED]

B2.8 • Endoscapes2023 • 4 • 24:58

Question: Based on the frame at 24:58, is it medically safe to proceed with continuing dissection toward clipping or cutting the cystic duct or artery? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. While the critical anatomical exposure and tissue plane prerequisites appear to be established, the actual hemostasis status, precise instrument position relative to the target structures, and energy safety cannot be verified. You must visually confirm that the cystic duct and artery are clearly isolated in a safe plane, that bleeding is fully controlled, that instruments are correctly positioned without threatening adjacent anatomy, and that energy devices are safely disengaged or positioned away from vulnerable tissues before applying clips or cutting. [VISUAL EVIDENCE NEEDED]

Rationale: The requested safety assessment relies on direct visual evidence at 24:58. The available annotations indicate that all three Critical View of Safety criteria—structures identified, hepatocystic triangle cleared, and gallbladder separated—are recorded as achieved with a unanimous score of 1.00, and the labeled structure classes include the calot triangle, cystic artery, cystic duct, cystic plate, and gallbladder. However, the provided metadata explicitly flags that hemostasis, actual instrument position, and energy safety are unlabeled and insufficient for a full visible safety judgment. Because clearing this step requires direct observation of tissue plane integrity, hemostasis, instrument placement, and energy safety to rule out injury, these critical factors cannot be verified from the provided metadata alone. [VISUAL EVIDENCE NEEDED]

B2.9 • Endoscapes2023 • 57 • 18:26

Question: Based on the frame at 18:26, is it medically safe to proceed with continuing dissection toward clipping or cutting the cystic duct or artery? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. Anatomical exposure and tissue plane are not yet established, as the critical structures remain unidentifiable and the hepatocystic triangle is uncleared. Hemostasis, instrument position, and energy safety are uncertain and cannot be verified. [VISUAL EVIDENCE NEEDED]

Rationale: The judgment relies on Endoscapes2023 annotations for video 57, clip b2_endoscapes_04 at 18:26. The available safety framework shows that structure identification and hepatocystic triangle clearance are both unmet (0.00), while gallbladder separation is fully met (1.00). Labeled classes confirm the presence of the cystic artery, cystic duct, cystic plate, and gallbladder, but these labels only provide names and do not describe appearance, condition, or spatial relationships. Furthermore, hemostasis, instrument position, and energy safety are explicitly noted as unlabeled factors insufficient for a complete safety judgment. Because the foundational safety prerequisites are incomplete and critical visual checks are missing, proceeding is unsafe. [VISUAL EVIDENCE NEEDED]

B2.10 • Endoscapes2023 • 74 • 8:02

Question: Based on the frame at 8:02, is it medically safe to proceed with continuing dissection toward clipping or cutting the cystic duct or artery? Answer yes/no, including anatomical exposure, tissue plane, hemostasis, instrument position, and energy safety uncertain and justify using visible evidence.

Answer: No—do not proceed without direct visual confirmation. While the anatomical exposure is recognized and the tissue plane has been cleared in the triangle, hemostasis, exact instrument position, and energy safety cannot be verified from the available data. You must directly observe that bleeding is fully controlled, the instrument tips are precisely positioned away from unintended structures, and energy devices are safely managed before making any cuts or clips. Without confirming these factors in real time, proceeding carries an unacceptable risk of injury. [VISUAL EVIDENCE NEEDED]

Rationale: The available annotations indicate that the Critical View of Safety prerequisites are partially met: structure identification shows partial achievement at 0.67, while hepatocystic triangle clearance and gallbladder separation are fully achieved at 1.00. Labeled structures include the calot triangle, cystic artery, cystic duct, cystic plate, and gallbladder. However, the provided labels do not cover hemostasis, actual instrument position, or energy safety, which are required to fully assess safety for clipping or cutting. Because these critical visible factors are absent from the annotations, a definitive safety judgment cannot be made without direct observation. [VISUAL EVIDENCE NEEDED]

B3 | Risk Ranking

9 QA pairs • *BernBypass70*: 3, *Endoscapes2023*: 3, *StrasBypass70*: 3

B3.1 • *BernBypass70* • BBP18 • 8:37-9:02 to 103:05-103:07

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: Securing hemostasis takes precedence, with ischemic injury as the secondary focus. The persistent, unresolved bleeding episodes across multiple surgical phases demand immediate intervention to prevent fluid loss or hemodynamic compromise. Ischemic injury remains unresolved and requires vigilant monitoring to ensure tissue perfusion is maintained and does not progress.

Rationale: The provided annotations document two distinct intraoperative adverse-event categories for this video: bleeding and ischemic injury. Bleeding is ranked first due to its highest recorded severity of 4, its high frequency (9 events), and its unresolved status across multiple critical phases, including gastric pouch creation, anastomotic reinforcement, and defect closure. Ischemic injury is ranked second with a lower recorded severity of 1 and a single event, but it remains unresolved during the gastrojejunal anastomosis phase. Because both remain unrectified, they carry clinical urgency, but bleeding's higher severity and multiplicity establish it as the primary risk.

B3.2 • *BernBypass70* • BBP19 • 10:20-10:34 to 69:16-69:37

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: Bleeding is the highest priority, followed by thermal injury, and then mechanical injury. Persistent bleeding requires immediate attention because it frequently occurs across multiple procedural steps and carries a significant risk that threatens hemodynamic stability and obscures the operative field. Thermal injury is next, as its moderate impact and unrectified status demand careful monitoring to prevent delayed tissue damage. Mechanical injury ranks lowest in priority, though it still requires vigilance to ensure no structural compromise develops.

Rationale: The ranking is derived from the `category_ranking_summary` and `event_records_with_timestamps` annotations. Bleeding is prioritized first because it has the highest maximum severity score of 4, occurs most frequently (7 events across multiple phases such as gastric pouch creation, gastrojejunal anastomosis, and jejunojejunal anastomosis), and remains unrectified. Thermal injury ranks second with a maximum severity score of 3, occurring once during the cleaning and coagulation phase, and is also unrectified. Mechanical injury ranks third with a maximum severity score of 1, occurring once during gastrojejunal anastomosis, and is unrectified. All three categories are marked as unrectified, so clinical priority is driven by the maximum severity and frequency recorded in the annotations.

B3.3 • *BernBypass70* • BBP20 • 8:13-9:55 to 107:31-113:41

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: The highest clinical priority is the ongoing risk of bleeding, which has manifested repeatedly throughout the procedure and reached the most severe level recorded. The next priority is the risk of anastomotic insufficiency, supported by documented instances of incomplete closure that remain unresolved.

Rationale: The ranking is derived directly from the intraoperative adverse event annotations for this procedure. The dataset records two distinct adverse event categories: bleeding and insufficient closure of the anastomosis. Bleeding is ranked first because it has the highest maximum severity level (reaching the highest point on the clinical severity scale) and comprises the majority of the recorded events across numerous surgical steps, with several instances marked as unresolved. Insufficient closure of the anastomosis is ranked second because it has a significantly lower maximum severity level (reaching only the lowest point on the scale) and fewer total occurrences, though it also remains unresolved. The ranking strictly follows the provided severity hierarchy and unresolved status of these specific recorded events.

B3.4 • Endoscapes2023 • 21 • 49:12

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: Incomplete clearance of the hepatocystic triangle takes the highest priority, followed by inadequate identification of the required anatomical structures, with incomplete separation of the gallbladder being the lowest priority among these prerequisites.

Rationale: The provided annotations include per-frame safety criterion achievement values: 0.00 for hepatocystic triangle clearance, 0.33 for structure identification, and 0.67 for gallbladder separation. In this scoring system, a lower value reflects less achieved safety and therefore carries a higher clinical priority for correction. Applying this inverse relationship yields the stated ranking. Because no discrete adverse events or other safety markers are supplied, the ranking is restricted to these three documented prerequisites.

B3.5 • Endoscapes2023 • 31 • 35:06

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: The highest priority is inadequate identification of the required anatomical structures. Next is incomplete clearance of the hepatocystic triangle. The gallbladder is fully separated, so it does not require further intervention.

Rationale: The ranking derives from the per-frame safety readiness labels provided for timestamp 35:06. The label for required structure identification shows a value of 0.00, indicating it has not been achieved and represents the most immediate safety barrier. The label for hepatocystic triangle clearance shows a value of 0.67, indicating partial achievement and a secondary concern. The label for gallbladder separation shows a value of 1.00, indicating complete achievement. Following the standard prerequisite hierarchy, structure identification must be resolved before addressing triangle clearance. The provided class presence labels confirm the relevant anatomical regions are tracked in the dataset but do not describe their visual state.

B3.6 • Endoscapes2023 • 94 • 12:20

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: The highest priority is the incomplete separation of the gallbladder from the surrounding tissue. The next priority is the incomplete identification of the required anatomical structures. Clearance of the hepatocystic triangle is adequate and does not represent an immediate safety concern. Clipping or cutting should not be performed until the anatomy is clearly defined and the gallbladder is sufficiently separated to confirm a safe operative field.

Rationale: The provided annotations supply three Critical View of Safety (CVS) criterion averages for this timestamp. The gallbladder separation average is 0.33, indicating non-achievement by a majority. The structure identification average is 0.67, indicating partial achievement with some annotator disagreement. The hepatocystic triangle clearance average is 1.00, indicating unanimous achievement. Per safety protocols, unmet or partially met prerequisites must be ranked by urgency, with lower average scores representing higher clinical priority. Consequently, gallbladder separation ranks highest, followed by structure identification. The triangle clearance is fully met and excluded from the risk ranking. The labeled structure classes confirm the anatomical context, but appearance and position cannot be inferred, so no visual claims are made. Expert consensus averages introduce slight uncertainty regarding the exact anatomical state, but the priority order remains clear.

B3.7 • StrasBypass70 • SBP08 • 3:53-4:32 to 63:28-65:01

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: The immediate clinical focus should be on controlling bleeding, as it presents the most frequent and clinically significant threat to the patient. Mechanical injury follows in priority, given its potential for direct structural damage and the need to reassess instrument trajectories. Finally, insufficient closure of the anastomosis requires attention, though it carries a comparatively lower immediate threat level. Each of these documented events remains unresolved, meaning they require active management to prevent downstream complications such as hemodynamic compromise, tissue perforation, or anastomotic leakage.

Rationale: The ranking is grounded in the six recorded intraoperative adverse-event labels. Bleeding is documented four times and reaches the highest recorded severity tier of 3, occurring both early in the case and later during mesenteric defect closure. Mechanical injury also reaches a severity tier of 3 but occurs only once later in the procedure. Insufficient closure of the anastomosis is recorded once with a lower severity tier of 1. All events carry an unresolved status, indicating persistent risk. Consequently, bleeding and mechanical injury are prioritized due to their higher severity and frequency, while insufficient closure is ranked lowest among the explicitly recorded risks. Because this assessment relies solely on categorical and severity metadata without visual confirmation of tissue planes or hemostatic status, the ranking reflects the documented event profile rather than real-time operative conditions.

B3.8 • StrasBypass70 • SBP63 • 19:38-20:15 to 103:37-104:09

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: Stabilizing blood loss must come first, as the active bleeding episodes carry the greatest clinical impact and appear repeatedly across multiple surgical steps. After achieving hemostasis, direct management of thermal tissue injury should be addressed.

Rationale: The provided annotations document two distinct intraoperative adverse-event categories: bleeding and thermal injury. Both categories contain events that remain unresolved throughout the documented span. Bleeding occurs six times across phases such as gastric pouch creation, anastomosis defect closure, and Petersen space closure, with a maximum recorded severity of 4. Thermal injury occurs once during gastric pouch creation with a maximum recorded severity of 3. Using the maximum supplied severity and unresolved status to establish priority, the bleeding events present the greatest immediate threat and rank highest, followed by the thermal injury.

B3.9 • StrasBypass70 • SBP69 • 6:48-7:19 to 71:33-72:21

Question: Rank the intraoperative risks from highest to lowest clinical priority.

Answer: Thermal injury is the highest priority, given its potential for significant tissue damage and unresolved nature. Bleeding ranks second, as it is a recurring issue throughout the operation that requires consistent attention to hemostasis. Mechanical injury is the lowest priority among the documented events, occurring less frequently and generally causing less immediate harm than the other two categories.

Rationale: The ranking is derived from the supplied intraoperative adverse-event annotations for this Roux-en-Y gastric bypass case. Thermal injury is prioritized first because its recorded maximum severity is the highest of all documented events (3 out of 5), it remains unresolved, and it occurs during the opening of the alimentary limb in the jejunojejunal anastomosis phase. Bleeding ranks second despite lower individual severity (1 out of 5) because it is the most frequent recorded event (12 distinct instances spanning multiple phases from gastric pouch creation through Petersen space closure), indicating a persistent challenge that demands vigilance. Mechanical injury ranks third, with only 3 recorded instances at the lowest severity level (1 out of 5), making it less clinically urgent than the thermal and bleeding events. All recorded events are noted as unresolved, further supporting the need to address the thermal injury first, followed by the frequent bleeding episodes, and then the occasional mechanical trauma.

B4 | Near-Miss Detection

6 QA pairs • BernBypass70: 3, StrasBypass70: 3

B4.1 • BernBypass70 • BBP35 • 18:31-19:41

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. This clip documents an actual bleeding event rather than a narrowly avoided injury. A near-miss implies that a harmful complication was prevented without occurring, whereas here bleeding has already happened and remains unresolved. The focus should be on immediate hemostasis and surgical correction rather than preventing an event that has already taken place.

Rationale: The provided metadata records an actual bleeding event with a severity level of 5 that has not been rectified, taking place during the retrogastric dissection step of gastric pouch creation. Because the annotations explicitly document an active adverse event rather than a narrowly avoided incident, this does not qualify as a near-miss. A near-miss classification requires evidence that a harmful complication was successfully averted, whereas the supplied data confirms that bleeding has already occurred and remains unresolved. The labels establish that this adverse event occurred, but do not describe its visual appearance or physical cause. [VISUAL EVIDENCE NEEDED]

B4.2 • BernBypass70 • BBP54 • 74:32-79:53

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. This clip documents an actual adverse event rather than a narrowly avoided injury. The clinical context indicates that an incomplete closure of the anastomosis occurred during the defect-closing step of the gastrojejunal anastomosis. Because the insufficiency manifested as a realized complication rather than a situation that was successfully averted, it does not qualify as a near miss. The focus should be on recognizing this as a concrete technical shortfall that requires immediate attention and correction, rather than treating it as a fortunate brush with harm.

Rationale: The provided annotations for video BBP54, clip b4_bern_02, contain a documented intraoperative adverse event category of 'Insufficient closure of anastomosis' with a severity level of 1 during the gastrojejunal anastomosis phase (step s19_gastrojejunal_defect_closing) at timestamp 74:32-79:53. Per B4 guidelines, a recorded adverse event is explicitly not a near miss; it is a realized complication. Therefore, the correct classification is 'no.' The answer distinguishes this actual event from a narrowly avoided injury, as instructed. Because the question asks what the clip shows but no clip was provided, visual verification is impossible. [VISUAL EVIDENCE NEEDED]

B4.3 • BernBypass70 • BBP58 • 71:31-72:44

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. This clip documents an actual ischemic injury that occurred during the procedure, rather than a narrowly avoided incident. A near-miss requires a situation where harm was almost caused but successfully prevented through timely correction or caution. Here, the tissue damage has already happened and was not rectified, meaning the event represents a realized complication rather than a close call. Coaching should focus

on recognizing early signs of compromised blood flow and adjusting technique to prevent progression of ischemia in the jejunal tissue.

Rationale: The annotation provides a recorded event category of "Ischemic injury" with a severity level of 1 during the jejunal separation phase (step s21_jejunal_section). The event scope limit explicitly states that the labels establish the adverse event occurred but do not describe its visual appearance or physical cause. Per hard rule 11, the severity level and rectification status are kept in the rationale. The presence of a documented adverse event category directly answers the B4 template by classifying it as a realized complication rather than a near miss. Because the question asks what the clip shows but no video was provided, visual confirmation of the event's appearance or mechanism cannot be verified. [VISUAL EVIDENCE NEEDED]

B4.4 • StrasBypass70 • SBP08 • 63:28-65:01

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. The provided documentation identifies this as a completed technical lapse—an incomplete closure of the anastomosis—rather than a narrowly avoided injury. A near miss occurs when a potentially harmful complication is prevented before it happens, but the event itself indicates that the closure was already insufficient. The procedure took place during the mesenteric defect closure phase and requires immediate intraoperative correction to prevent downstream complications, rather than representing a situation where harm was successfully averted.

Rationale: The question asks whether the clip shows a near-miss. The available annotations for the timestamp 63:28-65:01 explicitly classify the occurrence as a recorded_event_category of "Insufficient closure of anastomosis" with a recorded_event_severity of 1. According to the dataset schema and hard rule 4, a documented adverse event is categorically distinct from a near miss. The event is situated within the closure_mesenteric_defect operative phase and step s37_mesenteric_defect_closing, and is noted as not rectified. Because the event_scope_limit annotation confirms that the labels establish the occurrence but do not describe its visual appearance or physical cause, I cannot verify what the clip visually depicts. The judgment relies entirely on the explicit adverse-event classification. [VISUAL EVIDENCE NEEDED]

B4.5 • StrasBypass70 • SBP52 • 101:04-102:12

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. This instance represents a documented mechanical injury that actually occurred, not a narrowly avoided complication. A near-miss implies that harm was almost caused but successfully averted before tissue damage happened. In this case, the injury was confirmed and remained unresolved, meaning the harm was not prevented. Properly classifying this as an actual adverse event rather than a close call is essential for accurate safety reporting and targeted surgical coaching.

Rationale: The supplied annotations categorize this instance as a recorded mechanical injury with a severity of 3 that was not rectified. According to the dataset guidelines, a confirmed adverse event is distinct from a near-miss, which requires explicit documentation of a narrowly avoided injury. The question asks what the clip shows, but no visual content was provided to verify the event's appearance or context. [VISUAL EVIDENCE NEEDED]

B4.6 • StrasBypass70 • SBP56 • 32:27-35:00

Question: Does this clip show a near-miss event that could have resulted in patient harm?

Answer: No. This clip documents a thermal injury that has already occurred, rather than a narrowly avoided incident. A near-miss refers to a situation where a hazard was identified and corrected before it could cause harm to the patient. Here, the event is a confirmed adverse injury that was not rectified during the recorded span, meaning the harm has already happened rather than being prevented.

Rationale: The answer relies on the recorded_event_category label, which specifies a thermal injury, and the recorded_event_severity label, which assigns a level of 3. The recorded_event_rectified label indicates the event was not corrected, and the event_timestamp spans 32:27-35:00 within the jejunojejunal_anastomosis phase. Because the annotations explicitly log a confirmed adverse event rather than a near-miss category, it cannot be classified as a narrowly avoided injury. The event_scope_limit annotation confirms that the visual appearance and physical cause are not described in the metadata. [VISUAL EVIDENCE NEEDED]

B5 | Stop Point

9 QA pairs • *BernBypass70*: 3, *Endoscapes2023*: 3, *StrasBypass70*: 3

B5.1 • *BernBypass70* • BBP20 • 60:07-60:23

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm? Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: The case must be halted at 60:07-60:23. The immediate reason for this stop is the detection of an incomplete closure at the surgical connection during the testing phase. Continuing past this point carries a direct risk of an anastomotic leak, which can rapidly lead to severe intra-abdominal infection and sepsis. The structure at risk is the newly created anastomosis itself. You need to stop all advancement, thoroughly inspect the site, and revise your suturing technique to guarantee a completely watertight seal before proceeding.

Rationale: The requested template asks for a supervisory stop point based on the provided adverse event metadata. The available annotations identify a recorded event categorized as "Insufficient closure of anastomosis" occurring during the anastomosis_test phase, specifically at the s24_ink_injection step, with the timestamp recorded as 60:07-60:23. The severity is noted as level 1 and it is unrectified, indicating an unresolved technical deficiency that requires immediate attention to prevent an anastomotic leak. The annotations establish the occurrence of this safety concern but explicitly limit the context by stating they do not describe the visual appearance or physical cause of the event. Therefore, while the timestamp and clinical rationale for pausing the case are fully supported by the provided metadata, verifying the exact visual trigger and anatomical positioning requires direct frame review. [VISUAL EVIDENCE NEEDED]

B5.2 • *BernBypass70* • BBP42 • 37:50-38:05

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm? Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: You should pause the procedure at 37:50-38:05. The trigger for this intervention is the onset of a mechanical injury event during the current phase of the operation. Continuing without addressing this risks compounding trauma to the bowel wall and jeopardizing the integrity of the jejunojejunal anastomosis. Before you resume, completely release any traction or pressure on the tissue, visually confirm that the bowel wall is intact, and reassess your instrument grip and leverage. Only proceed once you are confident that the tissue is no longer under mechanical stress and the anastomotic site remains secure.

Rationale: The recommendation to pause is derived from the recorded mechanical injury event timestamped at 37:50-38:05, which carries the highest severity level and remains unrectified. The operative phase and step (jejunojejunal anastomosis, specifically alimentary limb measurement) establish that the small bowel and the newly created anastomosis are the structures at risk during this maneuver. A high-severity, unrectified mechanical injury at this stage demands an immediate stop to prevent perforation or ischemia. The coaching focuses on releasing traction and verifying tissue integrity, which directly addresses the mechanical stress implied by the event category. Because the provided labels confirm the event's occurrence and phase but do

not describe its visual appearance or physical cause, direct verification of the tissue state is required. [VISUAL EVIDENCE NEEDED]

B5.3 • BernBypass70 • BBP47 • 62:56-64:21

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?

Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: Pause the case at 62:56-64:21. The onset of an ischemic injury during the horizontal stapling phase triggers this stop. The intervention prevents further tissue compromise and potential necrosis within the gastric pouch. Immediate corrective coaching should focus on reassessing tissue perfusion before applying or releasing any stapling devices, ensuring that the tissue is not under excessive tension or compromised by prolonged instrument manipulation, and verifying that adequate blood supply remains intact before proceeding with the next stapler firing or closure step.

Rationale: The answer is grounded in the recorded_event_category (Ischemic injury) and event_timestamp (62:56-64:21), which establish the precise moment to pause. The operative_phase (gastric_pouch_creation) and operative_step (s6_horizontal_stapling) provide the anatomical and procedural context to identify the gastric pouch as the structure at risk and to formulate corrective coaching focused on perfusion and tissue handling during stapling. Because the event_scope_limit explicitly states that the labels do not describe the visual appearance or physical cause, the exact visible trigger cannot be verified from the provided data, requiring later VLM verification. [VISUAL EVIDENCE NEEDED]

B5.4 • Endoscopes2023 • 106 • 36:23

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?

Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: Proceeding with clipping or cutting right now carries a significant risk of damaging the bile ducts, so the operation needs to be halted at 36:23. The dissection has not yet achieved a fully safe anatomical view. You are at risk of inadvertently injuring the common bile duct or major hepatic vessels if you continue. Take a step back, re-identify the critical landmarks, and ensure the triangle of Calot is completely cleared before you attempt to clip or cut any structures. Do not advance until you have a definitive, unambiguous view of the cystic duct and artery entering only the gallbladder.

Rationale: The provided annotations for timestamp 36:23 include Critical View of Safety criterion scores. The hepatocystic triangle is recorded as fully cleared (1.00), and the cystic duct and artery are present in the labeled structures. However, structure identification is at 0.67 and gallbladder separation is at 0.33, indicating that the required anatomical exposure is incomplete and lacks unanimous expert consensus. Because the gallbladder has not been sufficiently separated from the liver bed and structure identification is not fully achieved, proceeding to clip or cut poses a direct risk to the biliary tree. This incomplete safety state at 36:23 serves as the supervisory stop trigger. [VISUAL EVIDENCE NEEDED]

B5.5 • Endoscapes2023 • 48 • 37:17

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?
Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: The procedure must be paused at 37:17. The stop is triggered by the current operative state, which shows that the fundamental anatomical landmarks necessary for safe dissection have not been clearly identified. Proceeding at this stage carries a high risk of inadvertently damaging the cystic duct or artery. To prevent this, you need to stop advancing and focus entirely on delineating the cystic duct from the common bile duct and common hepatic duct, ensuring you have a complete and unambiguous view of the structures within Calot's triangle. Only after those boundaries are definitively established should you consider applying clips or continuing the dissection.

Rationale: The timestamp 37:17 corresponds to an incomplete Critical View of Safety (CVS) state, specifically where the structures_identified criterion is at 0.00, hepatocystic_triangle_cleared is at 0.67, and gallbladder_separated is at 1.00. Following the conservative prerequisite order for CVS assessment, unmet structure identification is the highest priority concern. The labeled structures present include the calot_triangle and cystic_duct, confirming the relevant anatomical region is being dissected. Because the key ducts have not been definitively identified, proceeding poses a direct threat to the biliary tree and adjacent vasculature. The corrective guidance targets the clinical necessity of clearly defining the cystic duct from the common bile duct and common hepatic duct before any clipping or cutting. [VISUAL EVIDENCE NEEDED]

B5.6 • Endoscapes2023 • 82 • 17:04

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?
Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: At 17:04, the procedure requires an immediate pause because the current dissection state shows that the critical anatomical landmarks necessary for safe dissection have not yet been fully established. This incomplete safety state is the trigger for the stop. Proceeding with clipping or cutting the cystic duct or artery at this stage carries a significant risk of iatrogenic bile duct or vascular injury. The corrective coaching is to halt any further division or clipping and instead continue careful dissection to definitively identify the cystic duct and artery and completely clear the hepatocystic triangle of all surrounding tissue before proceeding.

Rationale: The requested stop point is anchored to the annotated timestamp 17:04. The provided annotations indicate an incomplete Critical View of Safety state at this moment: the required structures have not been clearly identified (cvs_structures_identified: 0.33), and the hepatocystic triangle has not been cleared (cvs_hepatocystic_triangle_cleared: 0.00). Although partial gallbladder separation is noted (cvs_gallbladder_separated: 0.67), the unmet foundational criteria create an unsafe window for further dissection or clipping. The labeled presence of the cystic duct, cystic artery, and operative tools confirms the relevant anatomical context, but the incomplete safety state dictates a supervisory halt to prevent injury. [VISUAL EVIDENCE NEEDED]

B5.7 • StrasBypass70 • SBP06 • 68:32-71:04

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?

Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: The risk of uncontrolled mesenteric hemorrhage requires an immediate stop at the 68:32-71:04 interval. The onset of a mechanical injury to the mesenteric tissue during the jejunal separation phase is the direct trigger for this halt. Continuing without intervention risks exacerbating vascular damage and causing significant hemorrhage from the exposed mesentery. At this moment, immediately withdraw sharp instruments or energy devices from the operative field, apply direct manual or instrument-based pressure to achieve hemostasis, and carefully evaluate the extent of the tissue breach. Once bleeding is controlled and the mesenteric defect is assessed, resume the mesenteric opening with a slower, more deliberate dissection technique, ensuring precise instrument placement and avoiding blind traction.

Rationale: The supervisory stop is grounded in the recorded intraoperative adverse event categorized as a mechanical injury with a severity level of 5 during the jejunal separation phase, specifically the s20 mesenteric opening step. The event timestamp 68:32-71:04 marks the exact window where this complication occurred and remains unrectified. A severity 5 classification indicates a major tissue compromise requiring immediate intervention to prevent catastrophic hemorrhage or further structural damage. The coaching directives focus on standard surgical management for mesenteric trauma: halting dissection, achieving hemostasis, and reassessing the tissue plane before proceeding. Because the provided annotations establish the occurrence and phase of this adverse event but explicitly limit description of its visual appearance or physical cause, the specific visual trigger and instrument positioning required to verify this event cannot be confirmed without frame-level review. [VISUAL EVIDENCE NEEDED]

B5.8 • StrasBypass70 • SBP08 • 63:28-65:01

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm?

Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: The procedure must be paused at 63:28-65:01. At this stage, the incomplete closure of the mesenteric defect is the primary concern, as an open defect directly exposes the adjacent bowel to internal herniation and potential strangulation. Immediately refocus your technique on securing the mesenteric edges; take deeper, more spaced sutures to eliminate tension, double-check that the tissue is fully approximated before tying, and do not advance until the closure is mechanically sound.

Rationale: The stop point is grounded in the adverse-event annotation indicating an insufficient closure of the anastomosis occurring during the closure_mesenteric_defect phase (step s37_mesenteric_defect_closing) at timestamp 63:28-65:01. The event is recorded as unrectified (severity 1), meaning the closure gap remains open and poses an immediate risk of bowel entrapment or ischemia if the operation continues. Coaching focuses on reinforcing the mesenteric closure technique to prevent internal herniation. Because the provided annotations establish the adverse event but explicitly limit their scope to metadata rather than visual appearance or physical cause, the exact visual trigger for the stop cannot be verified here. [VISUAL EVIDENCE NEEDED]

B5.9 • StrasBypass70 • SBP69 • 45:34-46:28

Question: At which timestamp should a supervisor pause the trainee to prevent possible patient harm? Explain: what visible event triggered the stop, what injury is being prevented, which structure is at risk, and what corrective coaching should be given.

Answer: A thermal injury has compromised tissue integrity, requiring an immediate stop at 45:34-46:28. The onset of this adverse event serves as the trigger for the pause. The injury being prevented is thermal necrosis or perforation of the bowel wall. The structure at risk is the alimentary limb during the jejunojejunal anastomosis phase. Corrective coaching should focus on ceasing energy application immediately, maintaining safe distance between the active instrument and adjacent tissue, and carefully verifying tissue plane integrity before continuing dissection or suturing. Consistent monitoring of energy device placement and gentle tissue retraction will minimize the risk of further thermal damage.

Rationale: The annotations record a thermal injury event occurring at 45:34-46:28 within the jejunojejunal anastomosis phase, specifically during the s31_alimentary_limb_opening step. The event is categorized as a thermal injury with a severity level of 3 and is marked as unrectified, indicating an active and unresolved safety concern that warrants immediate procedural interruption. Because the annotations confirm the adverse event but do not capture its visual appearance or physical cause, the stop point is grounded in the documented onset of the thermal injury rather than a directly observed visual cue. The at-risk anatomy corresponds to the alimentary limb and jejunojejunal anastomosis site, where thermal spread can compromise bowel integrity. Corrective coaching should focus on energy device safety and tissue distance during this specific step. Since the prompt requests an explanation of a visible trigger, direct visual verification is required to confirm the exact moment and appearance of the injury. [VISUAL EVIDENCE NEEDED]

B6 | Cause of Error

5 QA pairs • MultiBypass140: 5

B6.1 • MultiBypass140 • BBP01 • 23:43-23:44

Question: At which timestamp should a supervisor pause the trainee for coaching, and why?

Answer: A supervisor should pause the trainee at timestamp 23:43-23:44. During crura dissection within the gastric pouch creation phase, a thermal injury occurred and remains unresolved. Proceeding without addressing this injury risks expanding tissue damage and compromising the structural integrity of the operative field. The trainee needs to immediately halt energy application, assess the affected area, and implement appropriate measures to manage the thermal effect before proceeding further.

Rationale: The answer is grounded in the provided intraoperative adverse-event metadata. The `iae_category` is 'Thermal injury', the `iae_severity` is 3, and `iae_rectified` is False, indicating an unresolved event. The `event_timestamp` is 23:43-23:44, which aligns with the `operative_phase` 'gastric_pouch_creation' and `operative_step` 's4_crura_dissection'. A severity of 3 denotes a moderate-to-significant adverse event that requires immediate attention to prevent further harm, and because it has not been rectified, it serves as the clear trigger for a supervisory pause. All specific field names and numeric values are confined to the rationale as required.

B6.2 • MultiBypass140 • BBP20 • 31:37-31:54

Question: At which timestamp should a supervisor pause the trainee for coaching, and why?

Answer: A supervisor should pause the trainee at timestamp 31:37-31:54. This interval captures a significant, unresolved intraoperative bleeding event occurring precisely during the closure of the gastrojejunal defect. Active hemorrhage at this stage obscures the operative field and threatens the integrity of the anastomosis. The trainee must stop the current suturing maneuver, prioritize achieving definitive hemostasis, and only resume defect closure once the bleeding is fully controlled.

Rationale: The coaching pause is grounded in the recorded intraoperative adverse event (IAE) spanning 31:37-31:54. The available annotations specify the event category as bleeding, with a severity level of 5, and explicitly note that the event remains unrectified (`iae_rectified`: False). This adverse event occurs during the `gastrojejunal_anastomosis` phase, specifically within step `s19_gastrojejunal_defect_closing`. A high-severity, unresolved bleed during a critical closure step represents an immediate clinical priority that warrants supervisory intervention. The rationale for the pause focuses on halting the ongoing maneuver to address the hemorrhage before it compromises the surgical field or the anastomotic repair, directly translating the recorded event category and unrectified status into clinical urgency.

B6.3 • MultiBypass140 • BBP39 • 14:41-14:45

Question: At which timestamp should a supervisor pause the trainee for coaching, and why?

Answer: An immediate pause is required at 14:41-14:45. A mechanical injury occurring during the stapling phase of the gastrojejunal anastomosis represents a critical threat to tissue integrity at a highly vulnerable anatomical site. Stopping right now allows the team to thoroughly inspect the area for any unintended lacerations or bleeding, ensure that the staple line remains secure, and prevent a catastrophic leak or

hemorrhage. Proceeding without addressing this type of trauma during a complex anastomosis step puts the patient at severe risk, so halting the procedure is the only safe course of action until the area is fully evaluated and stabilized.

Rationale: The answer is grounded in the MultiBypass140 adverse event metadata for video BBP39. The `iae\_category` of 'Mechanical injury' combined with an `iae\_severity` of 5 (the highest level on the dataset scale) and `iae\_rectified` marked as False establishes an unmitigated, high-urgency safety breach. The `operative\_phase` (gastrojejunal_anastomosis) and `operative\_step` (s18_gastrojejunal_stapling) provide the necessary clinical context, as stapling involves creating a critical connection where mechanical trauma can quickly lead to leaks or bleeding. Because the adverse event timestamp explicitly spans 14:41-14:45, this is the designated stop point. No visual observation was needed to determine the pause location, as the structured safety metadata explicitly flags this interval.

B6.4 • MultiBypass140 • BBP54 • 74:32-79:53

Question: At which timestamp should a supervisor pause the trainee for coaching, and why?

Answer: A supervisor should pause the trainee at timestamp 74:32-79:53. This is because the procedure requires the immediate correction of an incomplete closure at the gastrojejunal anastomosis site. Leaving this defect unaddressed carries a significant risk of leakage and subsequent infection. The trainee needs to stop the current workflow and focus entirely on achieving a secure, watertight closure of the anastomosis before proceeding to the next phase of the operation.

Rationale: The answer is grounded in the provided intraoperative adverse-event annotations. The event category is 'Insufficient closure of anastomosis,' which occurred during the 'gastrojejunal_anastomosis' phase and specifically at the 's19_gastrojejunal_defect_closing' step. The event timestamp is 74:32-79:53, and it has not been rectified. Although the dataset assigns a severity level of 1, indicating a mild or initial-stage deviation, the unrectified status during a critical defect-closing step warrants a supervisory pause to ensure the anastomosis is completed securely before proceeding. The rationale relies strictly on these provided metadata fields without inferring visual details or physical mechanisms.

B6.5 • MultiBypass140 • BBP58 • 71:31-72:44

Question: At which timestamp should a supervisor pause the trainee for coaching, and why?

Answer: A pause is necessary at timestamp 71:31-72:44. During the jejunal separation phase, specifically the jejunal section step, an ischemic injury to the bowel has occurred and has not yet been corrected. Halting immediately is essential to evaluate vascular compromise and restore adequate blood flow before continuing the resection, thereby preventing tissue necrosis and ensuring the viability of the upcoming anastomosis. Reassess the mesenteric blood supply and adjust dissection technique to protect the remaining bowel segment.

Rationale: The answer is grounded in the MultiBypass140 annotations for video BBP58, clip b6_multibypass_04. The `iae\_category` label identifies an ischemic injury, which carries a high clinical priority due to the risk of tissue necrosis and anastomotic failure. The `iae\_severity` is recorded as 1 on a 1-5 scale, and `iae\_rectified` is False, indicating the issue persists and requires immediate attention. The event occurs during the `jejunal\_separation` phase and `s21\_jejunal\_section` step, providing the procedural context for the coaching pause. The `event\_timestamp` explicitly marks the interval 71:31-72:44 as the point to intervene. No

visual evidence is claimed; the guidance is derived strictly from the adverse-event metadata and its procedural context.

B7 | CVS Safety Check

5 QA pairs • Endoscapes2023: 5

B7.1 • Endoscapes2023 • 1 • 19:54

Question: Based on the frame at 19:54, is it medically safe to proceed with clipping or cutting the cystic duct or artery?

Answer: Not yet. The necessary anatomical structures have not been clearly identified, the hepatocystic triangle has not been cleared, and the gallbladder has not been sufficiently separated from the surrounding tissue. Without confirming this critical view of safety, clipping or cutting carries a high risk of injuring adjacent structures.

Rationale: The available annotations indicate the status of the three Critical View of Safety criteria at timestamp 19:54. The criterion for structures identified (`cvs_c1_structures_identified`), the criterion for hepatocystic triangle clearance (`cvs_c2_hepatocystic_triangle_cleared`), and the criterion for gallbladder separation (`cvs_c3_gallbladder_separated`) are all recorded at 0.00. On the provided 0-1 scale, a score of 0.00 indicates non-achievement of that criterion. Because all three prerequisites for safe dissection remain unmet, the procedure should not proceed to clipping or cutting.

B7.2 • Endoscapes2023 • 2 • 20:42

Question: Based on the frame at 20:42, is it medically safe to proceed with clipping or cutting the cystic duct or artery?

Answer: No, it is unsafe to proceed at this moment. The anatomical exposure required to safely clip or cut the cystic duct or artery is incomplete. The critical structures have not been fully identified, and the gallbladder has not been sufficiently separated from the surrounding tissue. Although the hepatocystic triangle has been cleared, advancing without complete structure identification and gallbladder separation carries a high risk of injuring nearby bile ducts or blood vessels. Continue dissection until all necessary anatomical landmarks are clearly exposed and the gallbladder is fully separated before attempting any clipping or cutting.

Rationale: The answer is derived from the three Critical View of Safety (CVS) criterion scores recorded for frame 20:42. The field `cvs_c1_structures_identified` is 0.67, indicating partial achievement and unmet structure identification. The field `cvs_c2_hepatocystic_triangle_cleared` is 1.00, indicating full achievement. The field `cvs_c3_gallbladder_separated` is 0.33, indicating non-achievement. Because all three criteria must be met to confirm safety, the incomplete identification of structures and lack of gallbladder separation mean the required safety prerequisites are not satisfied. The answer translates these partial and unmet criteria into direct clinical guidance without referencing the numeric values or annotation field names, adhering to the constraint that coaching and safety decisions must be stated plainly rather than as reports on hidden grading processes.

B7.3 • Endoscapes2023 • 22 • 28:47

Question: Based on the frame at 28:47, is it medically safe to proceed with clipping or cutting the cystic duct or artery?

Answer: Yes. All three critical prerequisites for safe clipping or cutting have been definitively met. The required anatomical structures are clearly identified, the hepatocystic triangle has been fully cleared of tissue, and the gallbladder is adequately separated from the liver bed. You can proceed with confidence.

Rationale: The safety determination relies strictly on the Endoscapes2023 Critical View of Safety criteria annotated for this frame. The three criteria—`cvs_c1_structures_identified`, `cvs_c2_hepatocystic_triangle_cleared`, and `cvs_c3_gallbladder_separated`—each received a unanimous expert score of 1.00, confirming full achievement across all safety prerequisites. With no unmet criteria present, proceeding is safe.

B7.4 • Endoscapes2023 • 3 • 10:36

Question: Based on the frame at 10:36, is it medically safe to proceed with clipping or cutting the cystic duct or artery?

Answer: Not yet. Safe clipping or cutting requires that all three critical view prerequisites be fully established before any structure is divided. The anatomical exposure and dissection needed to clearly identify the relevant structures, clear the hepatocystic triangle, and separate the gallbladder are currently only partially complete. You should pause and continue the dissection until full clarity and exposure are achieved across all three steps before proceeding.

Rationale: The safety determination relies on the three Critical View of Safety criterion values recorded for this frame: structures identified (0.67), hepatocystic triangle cleared (0.67), and gallbladder separated (0.67). In this framework, scores near 1 indicate full achievement, while intermediate values reflect partial achievement or lack of unanimous annotator agreement. Because all three prerequisites are only partially met, the anatomical landmarks have not yet been definitively established to a safe standard. Therefore, advancing to clipping or cutting is unsafe until full clearance and identification are confirmed.

B7.5 • Endoscapes2023 • 6 • 10:32

Question: Based on the frame at 10:32, is it medically safe to proceed with clipping or cutting the cystic duct or artery?

Answer: Not yet. Clear identification of the required anatomical structures is essential before any clipping or cutting is performed. At this point, the critical structures in the hepatocystic triangle have not been definitively exposed, and the dissection plane around the gallbladder has not been fully completed. Proceeding without confirming this exposure carries a high risk of injuring the bile ducts or major vessels. Continue dissecting carefully to establish the complete critical view before applying any clips or scissors.

Rationale: The decision is grounded in the provided Endoscapes2023 annotations for frame 10:32. The annotation `cvs_c1_structures_identified` shows a value of 0.00, meaning the critical structures are not yet achieved. The annotation `cvs_c2_hepatocystic_triangle_cleared` shows 0.67, indicating the triangle is only partially cleared. The annotation `cvs_c3_gallbladder_separated` shows 1.00, indicating full clearance of the gallbladder. Because the first two prerequisites for the critical view are unmet, the anatomical exposure is

incomplete. This directly supports a "Not yet" response, as proceeding without full identification and triangle clearance violates the safety standard required before clipping or cutting.